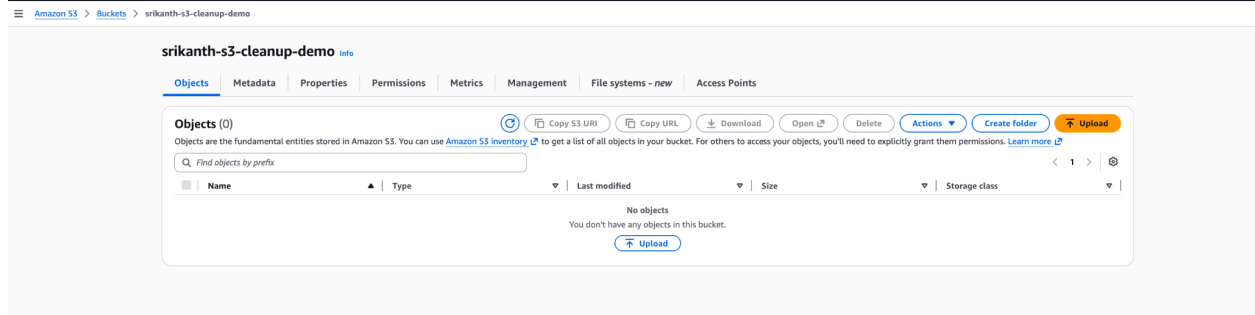


Task 1: Automated S3 Bucket Cleanup (Objects Older Than 30 Days)

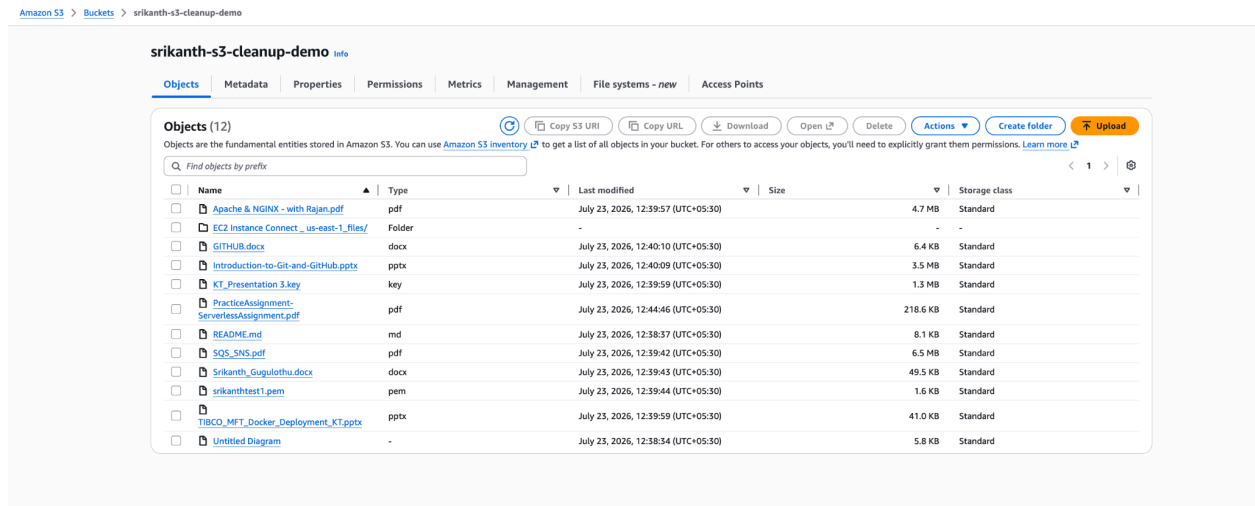
Step 1: Create S3 Bucket

Successfully created an Amazon S3 bucket named **srikanth-s3-cleanup-demo**.



Step 2: Upload Sample Files

Uploaded a few files and folders to the S3 bucket for testing.



Step 3: Create an IAM Role

Create an IAM role with the following configuration:

- **Trusted Entity:** AWS Service
- **Use Case:** Lambda
- **Permissions Policy:** **AWSLambdaBasicExecutionRole**
- **Role Name:** **S3BucketCleanupRole**

[Roles](#) > [Create role](#)

missions

[view, and create](#)

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and "+,=,_" characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: "-,=,_,@,/,|,[]#\$%^&*~'":

Step 1: Select trusted entities [Edit](#)

Trust policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "lambda.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }

```

Step 2: Add permissions [Edit](#)

Permissions policy summary

Policy name	Type	Attached as
AWSLambdaBasicExecutionRole	AWS managed	Permissions policy

Step 4: Add Inline Policy

Add an inline policy to the **S3BucketCleanupRole** IAM role with the following JSON policy.
Policy Name: **S3BucketCleanupPolicy**

[Roles](#) > [S3BucketCleanupRole](#) > [Edit policy](#)

permissions in
tCleanupPolicy

and save

Modify permissions in S3BucketCleanupPolicy

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

Visual | **JSON** | Actions

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "s3:ListBucket",
7       "Resource": "arn:aws:s3:::srinanth-s3-cleanup-demo"
8     },
9     {
10      "Effect": "Allow",
11      "Action": "s3:DeleteObject",
12      "Resource": "arn:aws:s3:::srinanth-s3-cleanup-demo/*"
13    }
14  ]
15 }

```

[+ Add new statement](#)

7:15 JSON

10008 of 10240 characters remaining

[Check for new access](#)

Security: 0 | Errors: 0 | Warnings: 0 | Suggestions: 0

[Cancel](#) [Next](#)

This inline policy grants the Lambda function permission to:

List the objects in the **srinanth-s3-cleanup-demo** bucket.

Delete objects from the **srinanth-s3-cleanup-demo** bucket.

nupRole

Policy S3BucketCleanupPolicy updated.

S3BucketCleanupRole [Info](#)

[Delete](#)

Allows Lambda functions to call AWS services on your behalf.

Summary [Edit](#)

Creation date
July 23, 2026, 12:49 (UTC+05:30)

Last activity
-

ARN
[arn:aws:iam::279473426275:role/S3BucketCleanupRole](#)

Maximum session duration
1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (2) [Info](#)

[Simulate](#) [Remove](#) [Add permissions](#)

Search

Filter by Type

All types

☐	Policy name ↗	Type	Attached entities
☐	AWSLambdaBasicExecutionRole	AWS managed	3
☐	S3BucketCleanupPolicy	Customer inline	0

▶ **Permissions boundary** (not set)

▼ **Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

Step 5: Create Lambda Function

Create an AWS Lambda function with the following configuration:

Function Name: S3BucketCleanup

Runtime: Python 3.12

Execution Role: S3BucketCleanup-role-b3j1ag1m

> [Functions](#) > S3BucketCleanup

S3BucketCleanup

Throttle Copy ARN Actions

Function overview [Info](#)

Diagram Template

+ Add trigger

+ Add destination

Function ARN: [arn:aws:lambda:us-east-1:279473426275:function:S3BucketCleanup](#)

Description: -

Last modified: 5 minutes ago

Export to Infrastructure Composer Download

Code Test Monitor **Configuration** Aliases Versions

Configuration

- General configuration
- Triggers
- Permissions**
- Destinations
- Function URL
- Environment variables
- Tags
- VPC
- RDS databases
- Monitoring and operations tools
- Concurrency and recursion detection
- Asynchronous invocation
- Code signing
- File systems

Execution role

Role name: [S3BucketCleanup-role-b3j1aglm](#)

View role document Edit

Resource summary

To view the resources and actions that your function has permission to access, choose a service.

Amazon CloudWatch Logs
3 actions 2 resources

By action By resource

Resource	Actions
arn:aws:logs:us-east-1:279473426275:*	Allow: logs:CreateLogGroup
arn:aws:logs:us-east-1:279473426275:log-group:/aws/lambda/S3BucketCleanup*	Allow: logs:CreateLogStream Allow: logs:PutLogEvents

Lambda obtained this information from the following policy statements:

- Managed policy AWSLambdaBasicExecutionRole-5c3d244a-dff8-4ebf-b946-121e71bf8a07, statement 0
- Managed policy AWSLambdaBasicExecutionRole-5c3d244a-dff8-4ebf-b946-121e71bf8a07, statement 1

Changed the code and successfully deployed it

Step 6: Create a Test Event

Create a test event to invoke the Lambda function with the following configuration:

- Event Name: **CleanupTest**
- Event JSON:



Code **Test** Monitor Configuration Aliases Versions

Test event [Info](#)

To invoke your function without saving an event, configure the JSON event, then choose Test.

CloudWatch Logs Live Tail Save Test

Test event action

Create new event Edit saved event

Invocation type

☒ Synchronous
Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

☐ Asynchronous
Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

CleanupTest

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

☒ Private
This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)

☐ Shareable
This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

Hello World

Event JSON

Format JSON

```
1 {  
2   |  
3 }
```

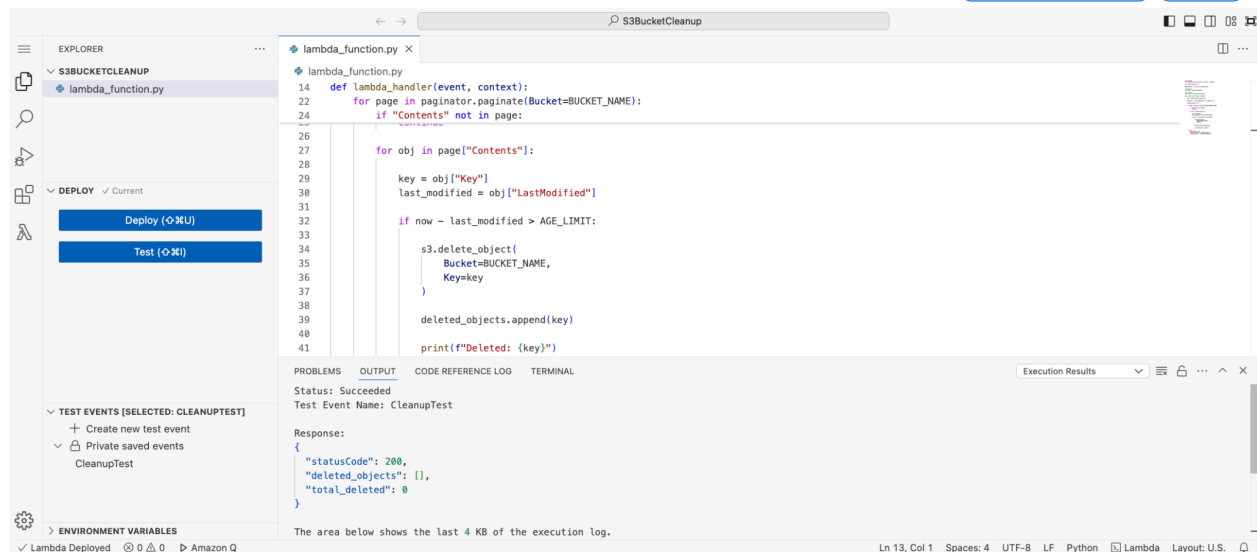
Step 7: Test the Lambda

After creating the test event, click the **Test** button to invoke the Lambda function.

The Lambda function executed successfully, and the test result showed a **Status: Succeeded**, indicating that the function ran without any errors.

Expected Result:

- **Test Event:** CleanupTest
- **Execution Status:** Succeeded
- **Response:** The Lambda function completed successfully and processed the S3 bucket cleanup logic.



When the timedelta(minutes=5)

```
{
  "statusCode": 200,
  "deleted_objects": [
    "Apache & NGINX - with Rajan.pdf",
    "EC2 Instance Connect _
us-east-1_files/65350ed04bbf4c6db33f8bb136e843b9023589d5c5d14f7590177a6810c657e5.js",
    "EC2 Instance Connect _ us-east-1_files/awsc-aperture-form-widget.js",
    "EC2 Instance Connect _ us-east-1_files/awsc-head.js",
    "EC2 Instance Connect _
us-east-1_files/e6d8172cdae34263bffeef6109514689d21943dc77be049d8b9b25bb1e37da3e0.js",
    "EC2 Instance Connect _ us-east-1_files/iFrame.html",
    "EC2 Instance Connect _ us-east-1_files/index(1).js",
    "EC2 Instance Connect _ us-east-1_files/index(2).js",
```

```

"EC2 Instance Connect _ us-east-1_files/index(3).js",
"EC2 Instance Connect _ us-east-1_files/index.css",
"EC2 Instance Connect _ us-east-1_files/index.js",
"EC2 Instance Connect _ us-east-1_files/isolatedIFrame.html",
"EC2 Instance Connect _ us-east-1_files/main.css",
"EC2 Instance Connect _ us-east-1_files/main.js",
"EC2 Instance Connect _ us-east-1_files/module.css",
"EC2 Instance Connect _ us-east-1_files/oauth",
"EC2 Instance Connect _ us-east-1_files/polaris-global-css(1).css",
"EC2 Instance Connect _ us-east-1_files/polaris-global-css.css",
"EC2 Instance Connect _ us-east-1_files/saved_resource.html",
"EC2 Instance Connect _ us-east-1_files/vendors.css",
"EC2 Instance Connect _ us-east-1_files/vendors.js",
"GITHUB.docx",
"Introduction-to-Git-and-GitHub.pptx",
"KT_Presentation 3.key",
"PracticeAssignment-ServerlessAssignment.pdf",
"README.md",
"SQS_SNS.pdf",
"Srikanth_Gugulothu.docx",
"TIBCO_MFT_Docker_Deployment_KT.pptx",
"Untitled Diagram",
"srikanthtest1.pem"
],
"total_deleted": 31
}

```

The area below shows the last 4 KB of the execution log.

Function Logs:

START RequestId: 86731df9-6fb5-453a-8c2c-67a724be91ab Version: \$LATEST

Deleted: Apache & NGINX - with Rajan.pdf

Deleted: EC2 Instance Connect _
us-east-1_files/65350ed04bbf4c6db33f8bb136e843b9023589d5c5d14f7590177a6810c657e5.js

Deleted: EC2 Instance Connect _ us-east-1_files/awsc-aperture-form-widget.js

Deleted: EC2 Instance Connect _ us-east-1_files/awsc-head.js

Deleted: EC2 Instance Connect _
us-east-1_files/e6d8172cdae34263bffe6109514689d21943dc77be049d8b9b25bb1e37da3e0.js

Deleted: EC2 Instance Connect _ us-east-1_files/iFrame.html

Deleted: EC2 Instance Connect _ us-east-1_files/index(1).js

Deleted: EC2 Instance Connect _ us-east-1_files/index(2).js

Deleted: EC2 Instance Connect _ us-east-1_files/index(3).js

```
Deleted: EC2 Instance Connect _ us-east-1_files/index.css
Deleted: EC2 Instance Connect _ us-east-1_files/index.js
Deleted: EC2 Instance Connect _ us-east-1_files/isolatedIFrame.html
Deleted: EC2 Instance Connect _ us-east-1_files/main.css
Deleted: EC2 Instance Connect _ us-east-1_files/main.js
Deleted: EC2 Instance Connect _ us-east-1_files/module.css
Deleted: EC2 Instance Connect _ us-east-1_files/oauth
Deleted: EC2 Instance Connect _ us-east-1_files/polaris-global-css(1).css
Deleted: EC2 Instance Connect _ us-east-1_files/polaris-global-css.css
Deleted: EC2 Instance Connect _ us-east-1_files/saved_resource.html
Deleted: EC2 Instance Connect _ us-east-1_files/vendors.css
Deleted: EC2 Instance Connect _ us-east-1_files/vendors.js
Deleted: GITHUB.docx
Deleted: Introduction-to-Git-and-GitHub.pptx
Deleted: KT_Presentation 3.key
Deleted: PracticeAssignment-ServerlessAssignment.pdf
Deleted: README.md
Deleted: SQS_SNS.pdf
Deleted: Srikanth_Gugulothu.docx
Deleted: TIBCO_MFT_Docker_Deployment_KT.pptx
Deleted: Untitled Diagram
Deleted: srikanthtest1.pem
END RequestId: 86731df9-6fb5-453a-8c2c-67a724be91ab
REPORT RequestId: 86731df9-6fb5-453a-8c2c-67a724be91ab Duration: 1748.27 ms Billed
Duration: 2224 ms Memory Size: 128 MB Max Memory Used: 96 MB Init Duration: 475.37
ms

Request ID: 86731df9-6fb5-453a-8c2c-67a724be91ab
```

Step 8: View CloudWatch Logs

After successfully testing the Lambda function, verify the execution logs in Amazon CloudWatch.

The CloudWatch log stream displays the Lambda execution details, including the start of the request, the objects deleted from the S3 bucket, and the execution summary.

Management > /aws/lambda/538bucketCleanup > 2026/07/23/[\$LATEST]d7e026cb04064f2ba40b36ac2a146b00

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp	Message
2026-07-23T07:56:01.585Z	Deleted: EC2 Instance Connect _ us-east-1_files/awsc-apture-form-widget.js
2026-07-23T07:56:01.625Z	Deleted: EC2 Instance Connect _ us-east-1_files/awsc-head.js
2026-07-23T07:56:01.665Z	Deleted: EC2 Instance Connect _ us-east-1_files/e6d8172cdae34263bffe6189514689a21943dc77be043d889b25b01e37a03e0.js
2026-07-23T07:56:01.687Z	Deleted: EC2 Instance Connect _ us-east-1_files/iFrame.html
2026-07-23T07:56:01.753Z	Deleted: EC2 Instance Connect _ us-east-1_files/index(1).js
2026-07-23T07:56:01.773Z	Deleted: EC2 Instance Connect _ us-east-1_files/index(2).js
2026-07-23T07:56:01.805Z	Deleted: EC2 Instance Connect _ us-east-1_files/index(3).js
2026-07-23T07:56:01.845Z	Deleted: EC2 Instance Connect _ us-east-1_files/index.css
2026-07-23T07:56:01.885Z	Deleted: EC2 Instance Connect _ us-east-1_files/index.js
2026-07-23T07:56:01.905Z	Deleted: EC2 Instance Connect _ us-east-1_files/isolatedIframe.html
2026-07-23T07:56:01.945Z	Deleted: EC2 Instance Connect _ us-east-1_files/main.css
2026-07-23T07:56:01.985Z	Deleted: EC2 Instance Connect _ us-east-1_files/main.js
2026-07-23T07:56:02.025Z	Deleted: EC2 Instance Connect _ us-east-1_files/module.css
2026-07-23T07:56:02.065Z	Deleted: EC2 Instance Connect _ us-east-1_files/oauth
2026-07-23T07:56:02.105Z	Deleted: EC2 Instance Connect _ us-east-1_files/polaris-global-css(1).css
2026-07-23T07:56:02.128Z	Deleted: EC2 Instance Connect _ us-east-1_files/polaris-global-css.css
2026-07-23T07:56:02.165Z	Deleted: EC2 Instance Connect _ us-east-1_files/saved_resource.html
2026-07-23T07:56:02.205Z	Deleted: EC2 Instance Connect _ us-east-1_files/vendors.css
2026-07-23T07:56:02.246Z	Deleted: EC2 Instance Connect _ us-east-1_files/vendors.js
2026-07-23T07:56:02.285Z	Deleted: GITHUB.docx
2026-07-23T07:56:02.308Z	Deleted: Introduction-to-Git-and-GitHub.pptx
2026-07-23T07:56:02.345Z	Deleted: KT_Presentation 3.key
2026-07-23T07:56:02.385Z	Deleted: PracticeAssignment-ServerlessAssignment.pdf
2026-07-23T07:56:02.406Z	Deleted: README.md
2026-07-23T07:56:02.445Z	Deleted: SQS_SNS.pdf
2026-07-23T07:56:02.485Z	Deleted: Srikanth_Gugulothu.docx
2026-07-23T07:56:02.525Z	Deleted: TIRCO_MFT_Docker_Deployment_KT.pptx
2026-07-23T07:56:02.545Z	Deleted: Untitled Diagram
2026-07-23T07:56:02.585Z	Deleted: srikanthtest1.pem

Back to top

Step 9: Verify S3 Bucket

After the Lambda function completes successfully, return to the S3 bucket and verified that the older objects have been removed.

srikanth-s3-cleanup-demo

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
3746148-DSAModule1RunningNotes.pdf	pdf	July 23, 2026, 13:25:27 (UTC+05:30)	58.9 KB	Standard